CNN Week-3 (2520090047 - M. Harshika)

1. AIM

To study network queueing behaviour, observe the effect of offered traffic on throughput and packet loss, examine the effect of different buffer sizes, capture and analyse UDP packets using Wireshark, and apply Little's Law to estimate average queueing delay.

2. SOFTWARE AND TOOLS REQUIRED

- Ubuntu on WSL
- iPerf3 for traffic generation and throughput measurement
- Linux Traffic Control (tc) and iproute2 for bottleneck configuration
- Wireshark / TShark for packet capture and analysis
- Linux network namespaces and virtual Ethernet interfaces

3. NETWORK TOPOLOGY

Three network namespaces were used: host1 as the sender, router as the bottleneck device, and host2 as the receiver. Virtual Ethernet pairs connected host1 to the router and the router to host2.

Node	Interface / role	IP address
Host1	Sender – veth-h1	10.0.1.1/24
Router	Sender-facing – veth-r1	10.0.1.2/24
Router	Receiver-facing – veth-r2	10.0.2.1/24
Host2	Receiver – veth-h2	10.0.2.2/24

```
harshika@siri:~$ sudo ip netns list
harshika@siri:~$ sudo ip netns add host1
harshika@siri:~$ sudo ip netns add router
harshika@siri:~$ sudo ip netns add host2
harshika@siri:~$ sudo ip netns list
host2
router
host1
```

4. PROCEDURE AND IMPLEMENTATION

STEP 1: SOFTWARE INSTALLATION AND VERIFICATION

The required tools were installed in Ubuntu and their versions were checked.

Command(s):

sudo apt update

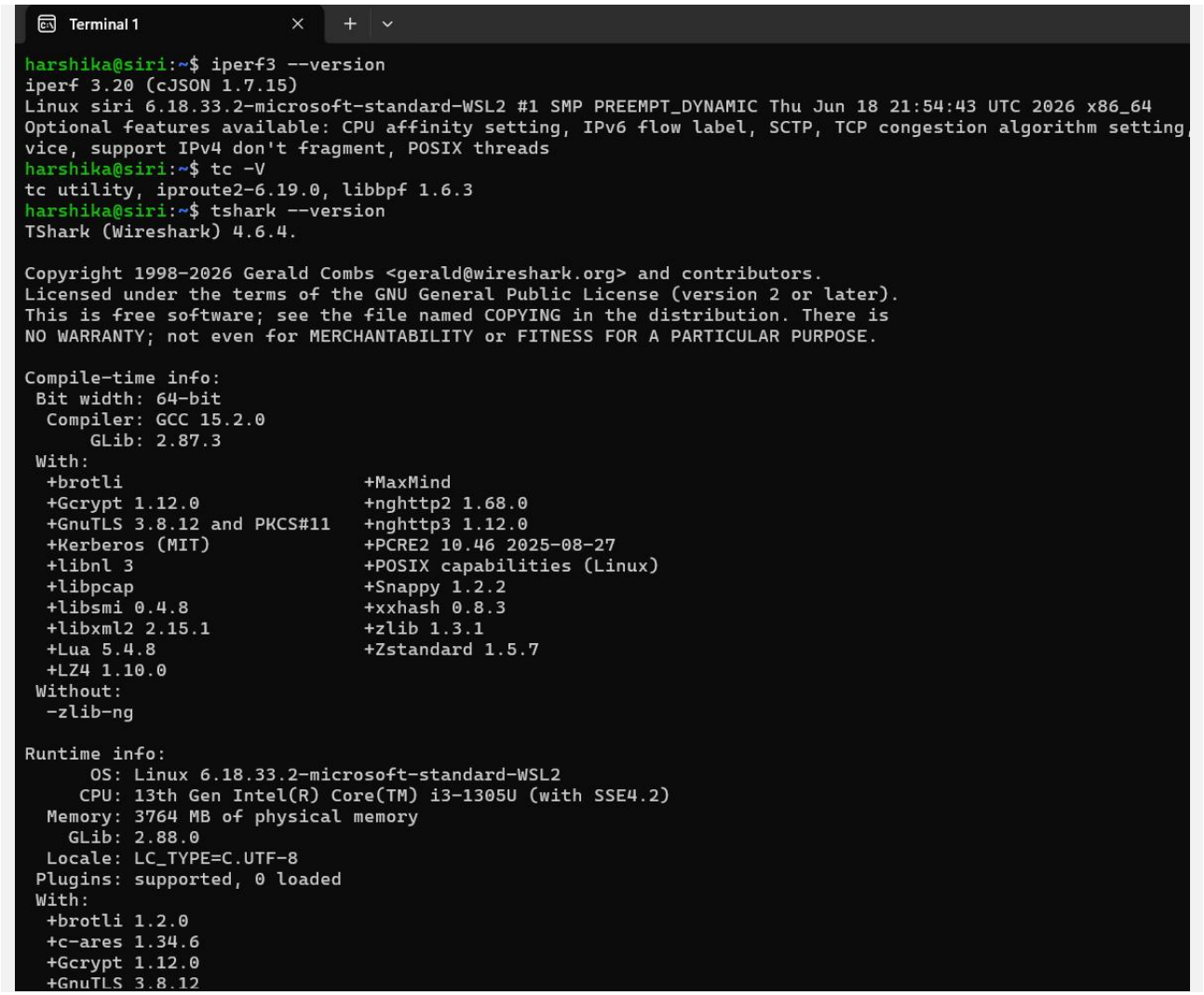
sudo apt install iperf3 iproute2 wireshark tshark

iperf3 --version

tc -V

tshark --version

Observation / Result: The required tools were installed and verified.



STEP 2: CREATE NETWORK NAMESPACES AND VIRTUAL LINKS

Three namespaces were created for the sender, router, and receiver. Virtual Ethernet pairs connected the namespaces.

Command(s):

sudo ip netns add host1

sudo ip netns add router

sudo ip netns add host2

sudo ip link add veth-h1 type veth peer name veth-r1

sudo ip link add veth-r2 type veth peer name veth-h2

Observation / Result: The three namespaces and virtual links were created. The veth interfaces were moved into their corresponding namespaces.

```
harshika@siri:~$ sudo ip link add veth-h1 type veth peer name veth-r1
harshika@siri:~$ sudo ip link set veth-h1 netns host1
harshika@siri:~$ sudo ip link set veth-r1 netns router
harshika@siri:~$ sudo ip link add veth-r2 type veth peer name veth-h2
harshika@siri:~$ sudo ip link set veth-r2 netns router
harshika@siri:~$ sudo ip link set veth-h2 netns host2
harshika@siri:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/24 dev veth-h1
harshika@siri:~$ sudo ip netns exec router ip addr add 10.0.1.2/24 dev veth-r1
harshika@siri:~$ sudo ip netns exec router ip addr add 10.0.2.1/24 dev veth-r2
harshika@siri:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/24 dev veth-h2
harshika@siri:~$ sudo ip netns exec host1 ip link set lo up
harshika@siri:~$ sudo ip netns exec host1 ip link set veth-h1 up
harshika@siri:~$ sudo ip netns exec router ip link set lo up
harshika@siri:~$ sudo ip netns exec router ip link set veth-r1 up
harshika@siri:~$ sudo ip netns exec router ip link set veth-r2 up
harshika@siri:~$ sudo ip netns exec host2 ip link set lo up
harshika@siri:~$ sudo ip netns exec host2 ip link set veth-h2 up
```

STEP 3: CONFIGURE IP ADDRESSES

IP addresses were assigned to the virtual interfaces and the interfaces were brought up.

Command(s):

host1 veth-h1: 10.0.1.1/24

router veth-r1: 10.0.1.2/24

router veth-r2: 10.0.2.1/24

host2 veth-h2: 10.0.2.2/24

Observation / Result: All interfaces were configured with the assigned IP addresses.

```
harshika@siri:~$ sudo ip netns exec host1 ip route add 10.0.2.0/24 via 10.0.1.2
harshika@siri:~$ sudo ip netns exec host2 ip route add 10.0.1.0/24 via 10.0.2.1
harshika@siri:~$ sudo ip netns exec host1 ping -c 4 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.704 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.065 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.076 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=0.053 ms

--- 10.0.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.053/0.224/0.704/0.276 ms
```

STEP 4: ENABLE FORWARDING AND CONFIGURE ROUTES

IP forwarding was enabled on the router and static routes were added on the sender and receiver.

Command(s):

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1

sudo ip netns exec host1 ip route add 10.0.2.0/24 via 10.0.1.2

sudo ip netns exec host2 ip route add 10.0.1.0/24 via 10.0.2.1

Observation / Result: IP forwarding and routing were configured successfully.

```
harshika@siri:~$ sudo ip netns exec router tc qdisc replace dev veth-r2 root tbf rate 5mbit burst 32kb limit 64kb
harshika@siri:~$ sudo ip netns exec router tc -s qdisc show dev veth-r2\
> ^C
harshika@siri:~$ sudo ip netns exec router tc -s qdisc show dev veth-r2
qdisc tbf 8001: root refcnt 7 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
```

STEP 5: TEST NETWORK CONNECTIVITY

The sender used ping to test connectivity with the receiver.

Command(s):

sudo ip netns exec host1 ping -c 4 10.0.2.2

Observation / Result: Four packets were transmitted and received, with 0% packet loss and average RTT of 0.224 ms.

```
harshika@siri:~$ sudo ip netns exec host1 ping -c 4 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.704 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.065 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.076 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=0.053 ms

--- 10.0.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.053/0.224/0.704/0.276 ms
```

STEP 6: CONFIGURE THE BOTTLENECK

A Token Bucket Filter (TBF) queueing discipline was applied to router interface veth-r2. The rate was limited to 5 Mbps with a 32 KB burst and 64 KB queue limit.

Command(s):

sudo ip netns exec router tc qdisc replace dev veth-r2 root tbf rate 5mbit burst 32kb limit 64kb

sudo ip netns exec router tc -s qdisc show dev veth-r2

Observation / Result: The outgoing interface was rate-limited to 5 Mbps with a 64 KB queue limit.

```
harshika@siri:~$ sudo ip netns exec router tc -s qdisc show dev veth-r2
qdisc tbf 8001: root refcnt 7 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
harshika@siri:~$ sudo ip netns exec host2 iperf3 -s
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
^Ciperf3: interrupt - the server has terminated by signal Interrupt(2)
harshika@siri:~$ sudo ip netns exec host2 pkill iperf3
harshika@siri:~$ sudo ip netns exec host2 iperf3 -s -D
```

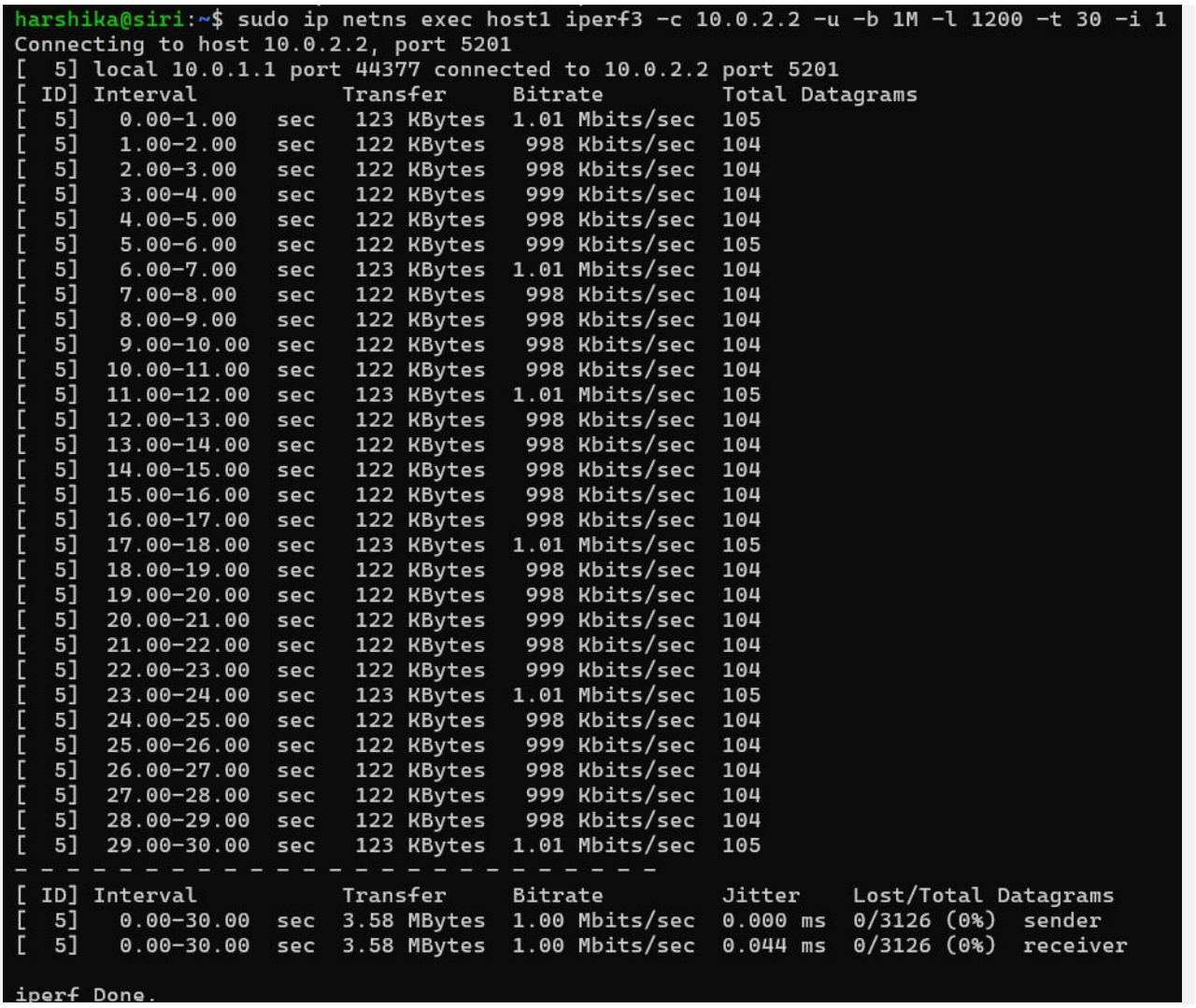
STEP 7: START THE IPERF3 RECEIVER

The iPerf3 server was started in the receiver namespace.

Command(s):

sudo ip netns exec host2 iperf3 -s -D

Observation / Result: The iPerf3 server was started in the background.



```
harshika@siri:~$ sudo ip netns exec host1 iperf3 -c 10.0.2.2 -u -b 1M -l 1200 -t 30 -i 1
Connecting to host 10.0.2.2, port 5201
[  5] local 10.0.1.1 port 44377 connected to 10.0.2.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   3.00-4.00   sec   122 KBytes   999 Kbits/sec  104
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   5.00-6.00   sec   122 KBytes   999 Kbits/sec  105
[  5]   6.00-7.00   sec   123 KBytes  1.01 Mbits/sec  104
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  11.00-12.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  12.00-13.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  13.00-14.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  14.00-15.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  15.00-16.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  16.00-17.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  17.00-18.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  18.00-19.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  19.00-20.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  20.00-21.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  21.00-22.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  22.00-23.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  23.00-24.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  24.00-25.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  25.00-26.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  26.00-27.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  27.00-28.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  28.00-29.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  29.00-30.00  sec   123 KBytes  1.01 Mbits/sec  105
- - - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  3.58 MBytes  1.00 Mbits/sec  0.000 ms  0/3126 (0%)  sender
[  5]   0.00-30.00  sec  3.58 MBytes  1.00 Mbits/sec  0.044 ms  0/3126 (0%)  receiver

iperf Done.
```

STEP 8: GENERATE UDP TRAFFIC AT DIFFERENT RATES

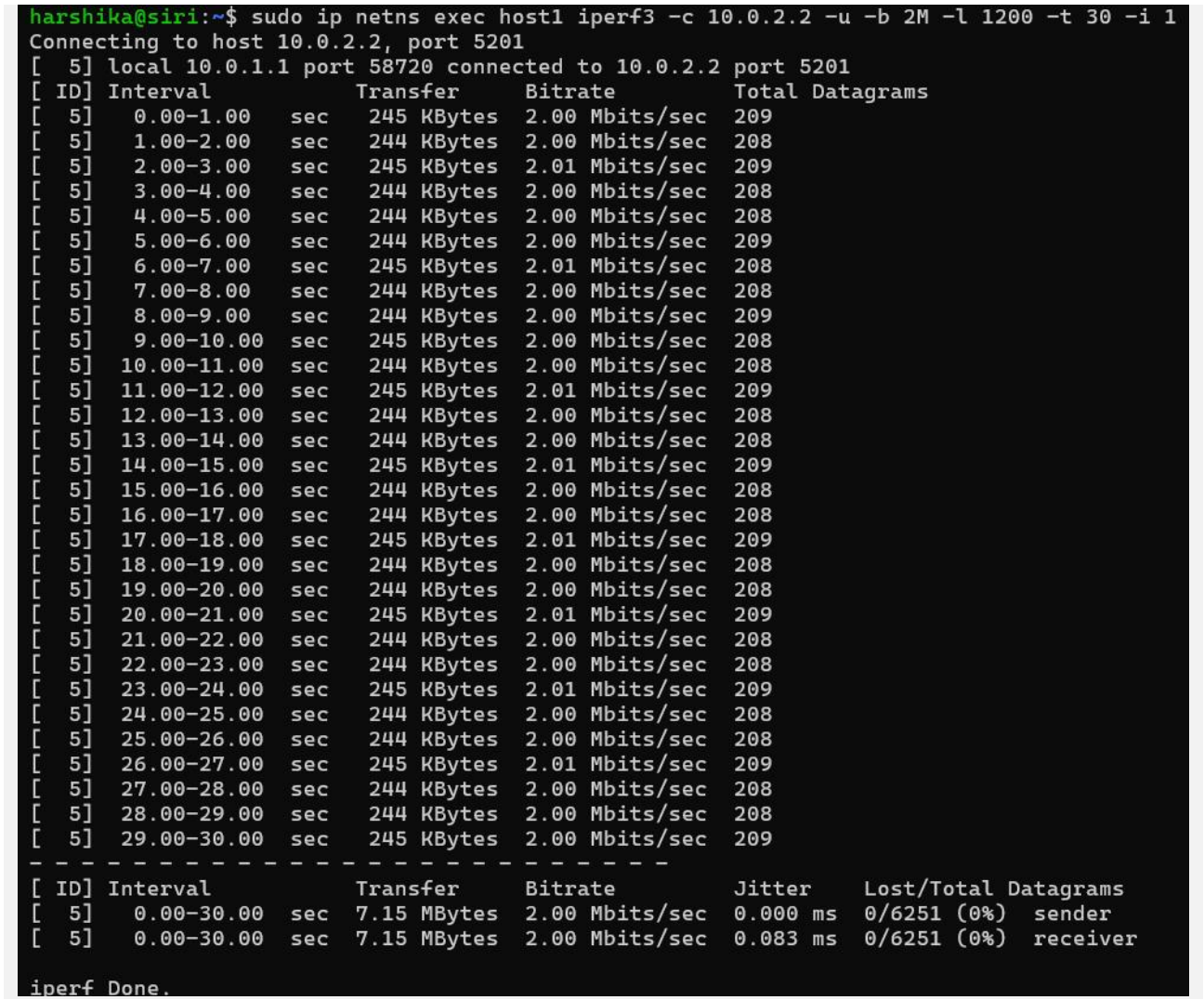
UDP traffic was generated for 30 seconds at 1, 2, 4, 5, 7, and 10 Mbps. Each test used 1200-byte datagrams and a one-second reporting interval.

Command(s):

sudo ip netns exec host1 iperf3 -c 10.0.2.2 -u -b 1M -l 1200 -t 30 -i 1

Repeat with -b 2M, -b 4M, -b 5M, -b 7M, and -b 10M.

Observation / Result: Receiver throughput approached the bottleneck capacity, while packet loss increased at higher offered rates.



5. QUEUEING CURVE OBSERVATIONS

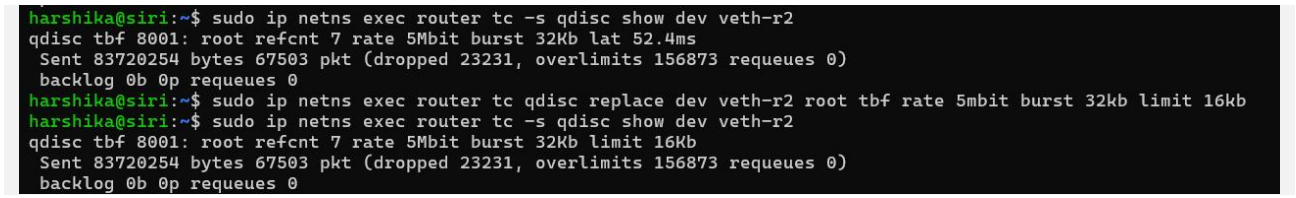
The offered UDP rate was increased while the bottleneck remained configured at 5 Mbps. The following values were recorded from the iPerf3 results.

Offered traffic	Receiver throughput	Packet loss
1 Mbps	1.00 Mbps	0%
2 Mbps	2.00 Mbps	0%
4 Mbps	4.00 Mbps	0%
5 Mbps	4.84 Mbps	2.9%
7 Mbps	4.84 Mbps	31%
10 Mbps	4.84 Mbps	51%

6. QUEUE STATISTICS

Queue statistics were inspected on the router's outgoing interface using: sudo ip netns exec router tc -s qdisc show dev veth-r2

Parameter	Recorded value
Sent bytes	83,720,254 bytes
Sent packets	67,503 packets
Dropped packets	23,231 (cumulative counter)
Overlimits	156,873 (cumulative counter)
Backlog after traffic stopped	0 bytes, 0 packets



7. BUFFER SIZE ANALYSIS

The TBF queue limit was changed to 16 KB, 64 KB, and 256 KB. A 7 Mbps UDP test was performed for each configuration while the bottleneck rate remained 5 Mbps.

sudo ip netns exec router tc qdisc replace dev veth-r2 root tbf rate 5mbit burst 32kb limit 16kb

sudo ip netns exec router tc qdisc replace dev veth-r2 root tbf rate 5mbit burst 32kb limit 64kb

sudo ip netns exec router tc qdisc replace dev veth-r2 root tbf rate 5mbit burst 32kb limit 256kb

Buffer limit	Offered rate	Receiver throughput	Packet loss	Jitter
16 KB	7 Mbps	4.84 Mbps	31%	0.613 ms
64 KB	7 Mbps	4.84 Mbps	31%	0.664 ms
256 KB	7 Mbps	4.84 Mbps	30%	0.638 ms

```
harshika@siri:~$ sudo ip netns exec router tc -s qdisc show dev veth-r2
qdisc tbf 8001: root refcnt 7 rate 5Mbit burst 32Kb lat 367ms
 Sent 121564823 bytes 98008 pkt (dropped 36515, overlimits 260980 requeues 0)
 backlog 0b 0p requeues 0
harshika@siri:~$ sudo ip netns exec host1 ping -c 10 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.541 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.074 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.062 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=0.069 ms
64 bytes from 10.0.2.2: icmp_seq=5 ttl=63 time=0.080 ms
64 bytes from 10.0.2.2: icmp_seq=6 ttl=63 time=0.078 ms
64 bytes from 10.0.2.2: icmp_seq=7 ttl=63 time=0.105 ms
64 bytes from 10.0.2.2: icmp_seq=8 ttl=63 time=0.080 ms
64 bytes from 10.0.2.2: icmp_seq=9 ttl=63 time=0.089 ms
64 bytes from 10.0.2.2: icmp_seq=10 ttl=63 time=0.239 ms

--- 10.0.2.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9222ms
rtt min/avg/max/mdev = 0.062/0.141/0.541/0.141 ms
```


```
harshika@siri:~$ sudo ip netns exec router tc -s qdisc show dev veth-r2
qdisc tbf 8001: root refcnt 7 rate 5Mbit burst 32Kb lat 367ms
 Sent 121564823 bytes 98008 pkt (dropped 36515, overlimits 260980 requeues 0)
 backlog 0b 0p requeues 0
harshika@siri:~$ sudo ip netns exec host1 ping -c 10 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.541 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.074 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.062 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=0.069 ms
64 bytes from 10.0.2.2: icmp_seq=5 ttl=63 time=0.080 ms
64 bytes from 10.0.2.2: icmp_seq=6 ttl=63 time=0.078 ms
64 bytes from 10.0.2.2: icmp_seq=7 ttl=63 time=0.105 ms
64 bytes from 10.0.2.2: icmp_seq=8 ttl=63 time=0.080 ms
64 bytes from 10.0.2.2: icmp_seq=9 ttl=63 time=0.089 ms
64 bytes from 10.0.2.2: icmp_seq=10 ttl=63 time=0.239 ms

--- 10.0.2.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9222ms
rtt min/avg/max/mdev = 0.062/0.141/0.541/0.141 ms
```


8. BASELINE PING AND PING UNDER LOAD

Baseline ping was measured before generating heavy traffic using: sudo ip netns exec host1 ping -c 10 10.0.2.2

Baseline parameter	**Observed value**
Packets transmitted / received	10 / 10
Packet loss	0%
Minimum RTT	0.062 ms
Average RTT	0.141 ms
Maximum RTT	0.541 ms

```
harshika@siri:~$ sudo ip netns exec router tshark -i veth-r2 -f "udp port 5201" -a duration:35 -w ~/Queueing_Analysis.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r2'
tshark: The file to which the capture would be saved ("/home/harshika/Queueing_Analysis.pcapng") could not be opened: Permission denied.
harshika@siri:~$ sudo ip netns exec router tshark -i veth-r2 -f "udp port 5201" -a duration:35 -w /tmp/Queueing_Analysis.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r2'

harshika@siri:~$ sudo ls -lh /tmp/Queueing_Analysis.pcapng
-rw------- 1 root root 380 Nov 13 03:45 /tmp/Queueing_Analysis.pcapng
harshika@siri:~$ sudo ip netns exec router tshark -i veth-r2 -f "udp port 5201" -a duration:40 -w /tmp/Queueing_Analysis.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r2'
15335
harshika@siri:~$ sudo ls -lh /tmp/Queueing_Analysis.pcapng
-rw------- 1 root root 19M Nov 13 03:48 /tmp/Queueing_Analysis.pcapng
harshika@siri:~$ sudo tshark -r /tmp/Queueing_Analysis.pcapng -c 10
```

Terminal 2

```
harshika@siri:~$ sudo ip netns exec host1 iperf3 -c 10.0.2.2 -u -b 7M -l 1200 -t 30 -i 1
[sudo: authenticate] Password:
Connecting to host 10.0.2.2, port 5201
[  5] local 10.0.1.1 port 40459 connected to 10.0.2.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   2.00-3.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   3.00-4.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   5.00-6.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   7.00-8.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   8.00-9.00   sec   853 KBytes  6.99 Mbits/sec  728
[  5]   9.00-10.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  10.00-11.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  11.00-12.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  12.00-13.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  13.00-14.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  14.00-15.00  sec   854 KBytes  6.99 Mbits/sec  729
[  5]  15.00-16.00  sec   853 KBytes  7.00 Mbits/sec  728
[  5]  16.00-17.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  17.00-18.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  18.00-19.00  sec   854 KBytes  6.99 Mbits/sec  729
[  5]  19.00-20.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  20.00-21.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  21.00-22.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  22.00-23.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  23.00-24.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  24.00-25.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  25.00-26.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  26.00-27.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  27.00-28.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  28.00-29.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  29.00-30.00  sec   854 KBytes  6.99 Mbits/sec  729
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  25.0 MBytes  7.00 Mbits/sec  0.000 ms  0/21876 (0%)  sender
[  5]   0.00-30.42  sec  17.5 MBytes  4.84 Mbits/sec  0.634 ms  6542/21874 (30%)  receiver

iperf Done.
```

9. LITTLE'S LAW CALCULATION

Little's Law is expressed as $L = \lambda \times W$, where L is the average number of packets in the queue, λ is the packet arrival rate in packets per second, and W is the average time spent in the queue in seconds.

During one active 7 Mbps test, ten backlog samples were recorded. The queued-packet counts were 210, 211, 211, 211, 211, 210, 211, 210, 210, and 211.

Average sampled queue occupancy: L = (210 + 211 + 211 + 211 + 211 + 210 + 211 + 210 + 210 + 211) / 10 = 210.6 packets

The iPerf3 sender reported 21,876 datagrams over 30 seconds. The offered packet rate was: λ offered = 21,876 / 30 = 729.2 packets/second

Using this offered rate as an approximation: W estimate = L / λ = 210.6 / 729.2 = 0.289 seconds ≈ 289 ms

Limitation: This is an estimate, not a strict validation of Little's Law. The 729.2 packets/second value is the offered sender rate, not a separately measured admitted rate at the queue. The ten backlog samples were collected during a saturated interval. A rigorous validation would use the admitted packet rate and a time-averaged queue occupancy measured over the same interval.

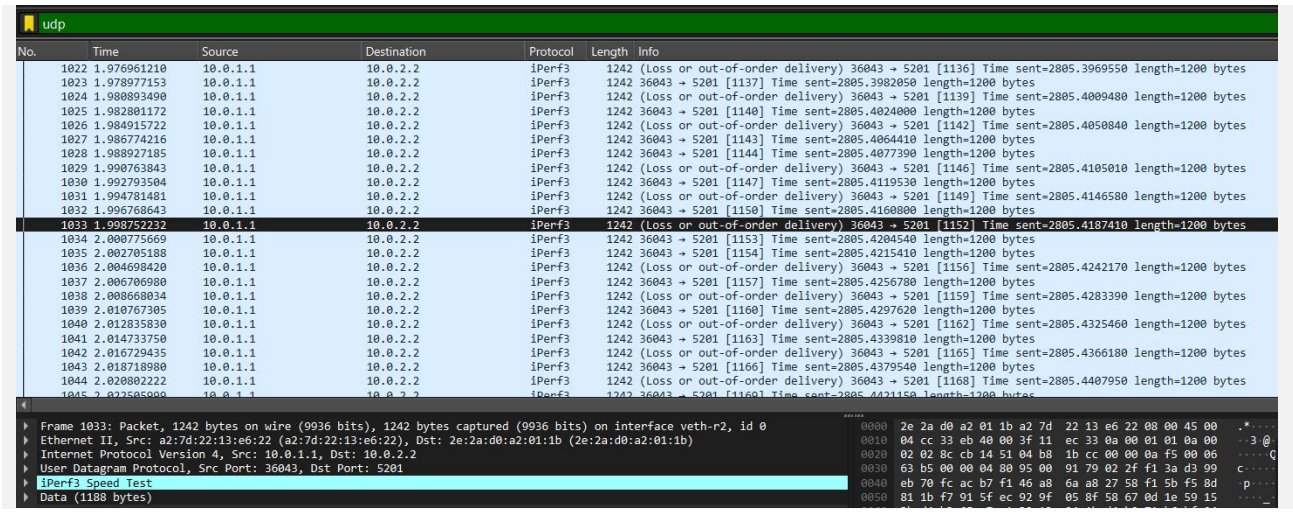
10. PACKET CAPTURE USING TSHARK

UDP traffic was captured on the router's outgoing interface veth-r2. The capture filter selected UDP port 5201, used by iPerf3.

sudo ip netns exec router tshark -i veth-r2 -f "udp port 5201" -a duration:40 -w /tmp/Queueing_Analysis.pcapng

The capture file was approximately 19 MB. The first packets were read back using: sudo tshark -r /tmp/Queueing_Analysis.pcapng -c 10

The packet list showed traffic from 10.0.1.1 to 10.0.2.2 using iPerf3 over UDP destination port 5201.



11. WIRESHARK I/O GRAPH ANALYSIS

The Queueing_Analysis.pcapng file was opened in Wireshark. Statistics → I/O Graphs was selected. The display filter udp.port == 5201 was applied and the interval was set to one second.

The graph showed a nearly steady stream of packets during the 30-second iPerf3 test and a fall in traffic at the end. The Y-axis was also changed to the Bits display unit for an additional view. The Bits unit should not be described as bits per second unless a rate unit is explicitly selected in Wireshark.



12. OVERALL OBSERVATIONS

• At 1, 2, and 4 Mbps offered traffic, receiver throughput closely followed the offered rate and no packet loss was observed.

• At and above the 5 Mbps bottleneck setting, receiver throughput remained close to 4.84 Mbps while packet loss increased.

• For a 7 Mbps offered load, the three buffer configurations produced approximately the same receiver throughput.

• The 256 KB buffer allowed a larger observed queue backlog than the smaller buffer settings.

• Ping delay and packet loss increased substantially during the heavy UDP load.

• Wireshark successfully captured the UDP packets and displayed their traffic pattern using I/O Graphs.